COMPLETE PRODUCT REQUIREMENTS DOCUMENT (PRD)

Project Name

Secure Spreadsheet Search Desktop

Executive Summary

Build a secure, offline-first, cross-platform desktop application that allows non-technical users to import spreadsheet files, search records quickly, and securely share searchable datasets with other users.

The application is intended for organizations that currently exchange Excel and CSV files through email, USB drives, shared folders, or messaging applications.

The application must eliminate common spreadsheet problems such as:

- Corrupted Excel files
- Multiple versions of the same file
- Difficult searching
- Accidental modification of records
- Data leakage
- Technical installation complexity

The application must work entirely offline and must never require cloud infrastructure.

Primary Objective

A user should be able to:

1. Install the application.
2. Import spreadsheet files.
3. Choose searchable columns.
4. Search records instantly.
5. Create a secure shareable package.
6. Send that package to another user.
7. Allow the recipient to search the data without needing the original spreadsheet.

User Types

User Type 1: Data Owner

Responsible for:

- Importing spreadsheet files
- Managing datasets
- Sharing datasets

Examples:

- HR staff
- Finance staff
- Administrative staff

User Type 2: Data Consumer

Responsible for:

- Receiving shared datasets
- Searching records

Examples:

- Customer support
- Branch office employees
- Managers

Supported Platforms

Must support:

- Windows 10+
- Windows 11
- macOS Intel
- macOS Apple Silicon
- Ubuntu Linux
- Major Linux distributions

Supported Spreadsheet Formats

Must support:

- XLSX
- XLS
- CSV
- TSV

Future Support:

- ODS

Architecture Requirements

The application must:

- Work completely offline
- Use local storage
- Never upload data
- Never require a server
- Never require an account

Recommended:

Frontend:

- React
- TypeScript

Desktop:

- Tauri

Database:

- SQLite

Encryption:

- SQLCipher

MASTER USE CASES

USE CASE 1

First Time Installation

Scenario:

User has never used the application before.

Steps:

1. User downloads installer.
2. User double-clicks installer.
3. User clicks Install.
4. Application installs.
5. Application launches automatically.

Expected Result:

User sees:

"Welcome"

Options:

- Import Spreadsheet
- Open Existing Dataset

No technical configuration required.

USE CASE 2

Import Spreadsheet

Scenario:

User has an Excel or CSV file.

Steps:

1. Open application.
2. Click Import Spreadsheet.
3. Select file.
4. Application reads file.
5. Application previews data.

Example:

EmployeeID	Name	Email
1001	John	john@test.com

Application displays columns.

User selects searchable columns.

Example:

☑ EmployeeID ☑ Name ☐ Email

User clicks Import.

Expected Result:

Dataset imported successfully.

Application displays:

Rows Imported: 10,250

USE CASE 3

Search Using Single Column

Scenario:

User wants to search by Employee ID.

Steps:

1. Open Search.
2. Select EmployeeID.

3. Enter:

1001

Expected Result:

Matching row displayed.

USE CASE 4

Search Using Multiple Columns

Scenario:

User searches:

John

Searchable Columns:

- Name
- Email

Expected Result:

Application returns all records containing John.

USE CASE 5

Partial Search

Scenario:

Search:

Joh

Expected Result:

Results include:

John Johnny Johnson

USE CASE 6

Case Insensitive Search

Scenario:

Database:

JOHN DOE

User searches:

john doe

Expected Result:

Record found.

USE CASE 7

Empty Search

Scenario:

User clicks Search without entering text.

Expected Result:

Application shows:

"Please enter search text."

USE CASE 8

No Results

Scenario:

Search:

999999999

Expected Result:

"No matching records found."

USE CASE 9

Import Very Large File

Scenario:

Spreadsheet contains:

100,000 rows

Steps:

Import file.

Expected Result:

Progress bar displayed.

Application remains responsive.

Import completes successfully.

USE CASE 10

Import Invalid Spreadsheet

Scenario:

User selects damaged file.

Expected Result:

"This file could not be imported."

No crash.

USE CASE 11

Create Shareable Bundle

Scenario:

User imported dataset.

Steps:

1. Open dataset.
2. Click Share Dataset.
3. Enter password.
4. Click Create Bundle.

Application creates:

dataset.companybundle

Expected Result:

Bundle exported successfully.

USE CASE 12

Send Bundle Via Email

Scenario:

User emails bundle.

Attachment:

dataset.companybundle

Expected Result:

Recipient receives bundle.

USE CASE 13

Recipient Already Has App

Scenario:

Recipient receives bundle.

Steps:

1. Double-click bundle.
2. Application opens.
3. Enter password.

Expected Result:

Dataset imported.

Search available immediately.

USE CASE 14

Recipient Does Not Have App

Scenario:

Recipient receives bundle.

Steps:

1. Double-click bundle.

Expected Result:

System displays:

"This file requires Secure Spreadsheet Search."

Options:

Install Application

After installation:

Bundle automatically imports.

USE CASE 15

Wrong Password

Scenario:

Recipient enters wrong password.

Expected Result:

Incorrect password.

Import blocked.

USE CASE 16

Corrupted Bundle

Scenario:

Bundle modified or damaged.

Expected Result:

Bundle validation fails.

Import rejected.

USE CASE 17

Reinstall Application

Scenario:

User reinstalls application.

Expected Result:

Previous datasets detected.

Application displays:

"We found your existing data."

Options:

- Restore
- Delete

USE CASE 18

Backup Dataset

Scenario:

User clicks Backup.

Expected Result:

Encrypted backup file generated.

USE CASE 19

Restore Backup

Scenario:

User imports backup.

Expected Result:

Dataset restored successfully.

USE CASE 20

Delete Dataset

Scenario:

User deletes dataset.

Expected Result:

Confirmation required.

Dataset removed.

SECURITY REQUIREMENTS

Security Requirement 1

No cloud services.

No telemetry.

No analytics containing user data.

Security Requirement 2

All local datasets encrypted.

Security Requirement 3

Shareable bundles encrypted.

Security Requirement 4

Passwords hashed.

Never stored in plaintext.

Security Requirement 5

Application logs must never contain:

- Spreadsheet contents
- Search values
- Sensitive company information

Security Requirement 6

Bundle tampering detection.

Digital signature validation.

Security Requirement 7

Optional:

Read-only mode.

Users can search but cannot export.

UI REQUIREMENTS

Home Screen

Buttons:

- Import Spreadsheet
- Search Data
- Manage Datasets
- Share Dataset
- Settings

Import Wizard

Step 1: Select File

Step 2: Preview Data

Step 3: Choose Searchable Columns

Step 4: Import

Search Screen

Components:

- Search box
- Column selector
- Search button
- Results table

Dataset Management Screen

Columns:

- Dataset Name
- Created Date
- Record Count
- Searchable Columns

Actions:

- Rename
- Delete
- Share
- Backup

PERFORMANCE REQUIREMENTS

Search:

50k records: <1 second

100k records: <2 seconds

500k records: <5 seconds

Import:

100k rows: <30 seconds

TEST CASES

Functional Tests

- Import XLSX
- Import XLS
- Import CSV
- Import TSV
- Search Exact Match
- Search Partial Match
- Search Multi Column
- Search No Results
- Delete Dataset
- Backup Dataset
- Restore Dataset
- Export Bundle
- Import Bundle

Security Tests

- Wrong Password
- Corrupted Bundle
- Modified Bundle
- Missing Signature
- Invalid Dataset

Installation Tests

- Fresh Install
- Upgrade Install
- Reinstall
- Uninstall Keep Data
- Uninstall Delete Data

Performance Tests

- 10k rows
- 50k rows
- 100k rows
- 500k rows

SUCCESS CRITERIA

A non-technical employee should be able to:

- Install the application
- Import a spreadsheet
- Select searchable columns
- Search records instantly
- Share data securely
- Receive shared data
- Use the application entirely offline

without understanding:

- Databases
- APIs
- Servers
- Cloud infrastructure
- Programming
- Command-line tools

The experience should feel as simple as opening a PDF file while providing enterprise-grade security for sensitive company data.